

Damilola Ogunleye

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EDUCATION

- University College London** London, UK
• *Master of Engineering in Robotics and Artificial Intelligence* 2023 - 2027
Second-Year Results: 84.27% (1st Class)
Modules: Object-Oriented Programming, Deep Learning for Robotics and Artificial Intelligence
- St Olav VGS** Stavanger, Norway
• *International Baccalaureate Diploma Programme* 2021 - 2023
Grade: 41/45 Maths AA HL: 7 Physics HL: 7

SKILLS SUMMARY

- Languages & Tools:** Python, PyTorch, MATLAB, LangGraph, C, scikit-learn, Git, Fusion360, PyBullet

EXPERIENCE

- National Center for Artificial Intelligence and Robotics** Abuja, Nigeria
• *Robotics Intern* August 2025 - September 2025
 - Agricultural Robot Prototype:** Prototyped an agricultural robot with Wi-Fi monitoring, developing seed-dispensing and furrow-opening mechanisms. Implemented GPS-IMU-Encoder sensor fusion with a Kalman filter for accurate navigation.
 - Impact:** Enhanced planting efficiency and harvest outcomes by designing a method for increased precision and high throughput compared to traditional farming practices in rural Nigeria.
- Cisco (Outshift by Cisco)** London, UK
• *Agentic AI Engineering Intern* April 2025 - July 2025
 - Agentic Workflow:** Designed and deployed a multi-agent orchestration pipeline using LangChain and LangGraph to coordinate LLM-driven agents for CAD design generation, iterative refinement, and automated 3D printing.

PROJECTS

- UKSEDS Olympus Rover Trials (Advanced Stream) – Perception, Navigation and Simulation Lead (November 2025 - Present):** Leading the development of a ROS2-based rover autonomy stack using RGB-D/LiDAR SLAM, A* planning, and MPPI control. Currently, optimising performance for Jetson/Raspberry Pi deployment via Gazebo simulation in a cross-functional team.
- Reinforcement Learning Control of Efficient Soft Exoskeletons (September 2025 - Present):** Designing a soft exoskeleton using twisted string actuators (TSAs). Developing a MuJoCo-based biomechanical simulation and exploring reinforcement learning controllers (PPO/SAC) for adaptive assistive torque generation.
- Multimodal AI Literacy Agent – Lead Deep Learning Researcher (September 2025 - December 2025):** Led a cross-functional team to architect a multimodal AI agent, integrating a fine-tuned LLM (RAG and dual-dataset fine-tuning) with a YOLO/CNN vision pipeline for real-time interaction using PyTorch. Benchmarked performance against state-of-the-art commercial agents.
- Secure Real-Time Distributed Flocking on ESP32 (FreeRTOS) (September 2025 - December 2025):** Designed and evaluated a secure real-time distributed flocking system on ESP32 using FreeRTOS, implementing deterministic task scheduling and decentralised Reynolds flocking over LoRa; built an authenticated wireless protocol with replay protection and analysed jitter, CPU usage, and communication reliability under adversarial conditions.
- UKSEDS Olympus Rover Trials (Basic Stream) – Robotics Engineer (November 2024 - July 2025):** Led the design and construction of a tracked Martian rover, integrating autonomous QR code detection (MicroPython) and a custom mobile interface for teleoperation across simulated terrain. Secured the ‘Best Innovation’ award and 2nd place overall.
- A Stochastic Simulation Framework for Robotic Grasp Stability Classification. (October 2024 - December 2024):** Engineered an Object-Oriented PyBullet simulation to stochastically generate grasping datasets, extracting kinematic and contact features to train PyTorch MLPs and Scikit-learn SVMs, achieving 79% accuracy in predicting grasp stability.
- Google BGN Hackathon — 1st Place & People’s Choice (October 2024):** Engineered a Python backend using Google Cloud Vision AI and Gemini API for real-time object detection and contextual retrieval, automating the image processing pipeline via GCP Cloud Storage Triggers.

HONORS AND AWARDS

- Best Innovation Award & 2nd Place Overall – UKSEDS Olympus Rover Trials 2025
- 1st Place – Google BGN Hackathon (Spotlight’ By Gemi9), October 2024

EXTRACURRICULARS

- Music** London, UK
• *Clarinetist – All-Souls Orchestra; Former member of Rogaland Regional Youth Symphony Orchestra.*
- Campus Societies** London, UK
• *Jazz Society, Computer Science Society, AI Society.*